# DANIEL CORRALES ALONSO

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## ABOUT ME

Predoctoral researcher at the Institute of Mathematical Sciences (ICMAT-CSIC) working on probabilistic machine learning and Bayesian methods for the design of model adaptation techniques, robustness to adversarial attacks and uncertainty-based decision making. I hold a degree in Mathematical Engineering from Universidad Complutense de Madrid and an MSc in Machine Learning for Health from Universidad Carlos III de Madrid. I strive to contribute to technology that is more interpretable, transparent, and fair.

## EDUCATION

**PhD at Mathematics Department** *Sept 2024 - Present*
Department of Mathematics, School of Sciences, Universidad Autónoma de Madrid, Spain.

**MSc in Machine Learning for Health** *Sept 2023 - Jul 2024*
Graduate School of Engineering and Basic Sciences, Universidad Carlos III de Madrid, Spain.

**BSc in Mathematical Engineering** *Sep 2018 - Mar 2023*
Department of Mathematical Sciences, Universidad Complutense de Madrid, Spain.

## PROFESSIONAL EXPERIENCE

**Department of Statistical Science, Duke University.** *Feb 2026 - May 2026*
*Research Stay* *Durham, North Carolina, US*

Visitor under the supervision of Professor David Banks. Worked on using Natural Language Processing to infer intent on foreign interference efforts during the US elections.

**Institute of Mathematical Sciences (ICMAT-CSIC)** *Jan 2024 - Sept 2024*
*Research Technician* *Madrid, Spain*

Research technician working in the ONCOSCREEN project funded by the European Union's Horizon Europe research and innovation programme. Development of colorectal cancer risk assessment tools and modelling of screening strategies.

**Institute of Mathematical Sciences (ICMAT-CSIC)** *Mar 2023 - Sep 2023*
*CSIC Scholarship for Introduction to Research. JAE Intro Program* *Madrid, Spain*

Research internship working on a project to analyse colorectal cancer risk in the European population and the performance of screening methods using Bayesian networks and inference diagrams. Developed under the supervision of Professor David Ríos-Insua.

**Basque Center for Applied Mathematics (BCAM)** *Jul 2022 - Sept 2022*
*Research Intern* *Bilbao, Spain*

Developed my B.Sc. final year project on the analysis, guarantees and implementation of the Minimax Risk Classifier algorithm under the supervision of Professor Santiago Mazuelas.

**Warwick Mathematics Institute, University of Warwick** *Sep 2021 - Jul 2022*
*Erasmus Scholarship Exchange Student Program* *Coventry, UK*

## PUBLICATIONS

1. Corrales, D., Insua, D. R. (2025, June). *Incentivising Personalised Colorectal Cancer Screening: an Adversarial Risk Analysis Approach.* In 2025 IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS) (pp. 307-310). IEEE. doi:10.1109/CBMS65348.2025.00068
2. Corrales, D., Insua, D. R., González, M. J. (2025). *A decision support model for colorectal cancer screening.* Computers in Biology and Medicine, 196, 110755. doi:10.1016/j.compbiomed.2025.110755
3. Corrales, D., Santos-Lozano, A., López-Ortiz, S., Lucia, A., Insua, D. R. (2024). *Colorectal cancer risk mapping through Bayesian networks.* Computer Methods and Programs in Biomedicine, 257, 108407. doi:10.1016/j.cmpb.2024.108407

## SEMINARS AND CONFERENCES

- **XLII National Congress on Statistics and Operational Research.** *Univ. Santiago de Compostela, Sep 2026*
  Talk: *A probabilistic framwework for test time adaptation.*
- **International Society for Bayesian Analysis World Meeting.** *Nagoya, Japan, June 2026*
  Poster: *Bayesian online test time adaptation: a general framework.*
- **5th Spanish Young Statisticians and Operational Researchers Meeting.** *Univ. de Sevilla, Nov 2025*
  Talk: *Designing incentives for colorectal cancer screening programs using adversarial risk analysis.*
- **Workshop: AI x Mathematics Camp 2025.** *Feishu China, Shanghai, Nov 2025*
- **Workshop: Accelerating statistical inference and experimental design with machine learning.** *Isaac Newton Institute for Mathematical Sciences, Cambridge, Jun 2025*
- **IEEE 38th Computer Based Medical Systems.** *Univ. Politécnica de Madrid, Jun 2025*
  Talk: *Incentivising personalised colorectal cancer screening: an adversarial risk analysis approach.*
- **XLI National Congress on Statistics and Operational Research.** *Univ. de Lleida, Jun 2025*
  Talk: *Modelling cancer screening strategies using Bayesian decision analysis.*
- **I Iberian Conference on MCDM/A.** *Universidade de Coimbra, May 2025*
  Talk: *Leveraging MCDA for colorectal cancer screening strategies.*
- **IV Jornadas Científicas de la PTI+ Salud Global.** *Univ. de Valencia, Oct 2024*
  Talk: *A decision analysis model for colorectal cancer screening.*
- **INFORMS Advances in Decision Analysis Conference.** *Aalto University, Jul 2024*
  Talk: *A decision analysis model for colorectal cancer screening.*
- **4th Spanish Young Statisticians and Operational Researchers Meeting.** *Univ. de Santiago de Compostela, Jun 2024*
  Talk: *A decision analysis model for colorectal cancer screening.*
- **First International Workshop on Bayesian Statistics.** *Univ. Carlos III de Madrid, Jun 2024*
  Talk: *A decision analysis model for colorectal cancer screening.*
- **Seminar on Multidimensional Security.** *Univ. Complutense de Madrid, Nov 2023*
  Talk: *Techniques towards explainability in machine learning models.*
- **Workshop: The many challenges of Artificial Intelligence.** *CSIC, Madrid, Nov 2023*
  Poster: *Colorectal cancer risk mapping through Bayesian Networks.*
- **AIHUB-CSIC Summer School.** *CSIC, Barcelona, Jul 2023*
  Workshop: *Machine learning hackathon.*

## MEDIA COVERAGE

**Un nuevo modelo matemático permite mejorar la detección de cáncer colorrectal**, *Actualidad del CSIC*, September 17th, 2025

## ACTIVITIES

- **UNU-INWEH x Feishu International Hackathon "AI + Mathematics + Sustainability"** *Nov 2025*
- **I DataFest Spain IKEA-UNAV** *Apr 2023*
- **Workshop: ExperimentaMates+ODS** *Nov 2022*
- **Accenture-UCM Modelling Competition Winner** *Nov 2021*

## SKILLS

**Programming**
Python (scikit-learn, tensorflow, pytorch, pgmpy, pymc), MATLAB, R.
**Languages**
Spanish (native), English (fluent, Cambridge University certification), German (studying for B1-level).